

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
6th Semester of B. Pharm. Examination
University Theory Examination May 2016
PH309 Pharmacognosy & Phytochemistry-IV

Date: 05.05.16, Thursday Time: 10:00 am to 1:00 pm Maximum Marks: 80

Instructions:

1. Figures to right indicate marks.
2. Section wise separate answer book to be used.
3. Draw neat sketches wherever necessary.

SECTION- I

Q.1 Attempt any TWO of the following:

- A** Define Alkaloids. Classify alkaloids according to the heterocyclic ring present in their structure giving examples in each class. Explain its properties and write about general chemical tests for detection of alkaloids. **10**
- B** Write a note on Purine alkaloids. Explain the Pharmacognosy of Coffee and Tea. **10**
- C** Write chemistry, isolation and estimation of Quinine and Caffeine. **10**

Q.2 Attempt any TWO of the following:

- A**
1. Explain the cultivation and preparation of Ergot and enlist its chemical constituents. **10**
 2. Differentiate- Rio and Cartegena Ipecac.
- B**
1. Write a note on Cinchona and its adulterants. **10**
 2. Draw the well labeled diagram of Transverse section of Vasaka leaf.
- C** Write a short note on Natural colorants and plant sweetening agents. **10**

SECTION- II

Q.3 Attempt any TWO of the following:

- A** 1. Contrast and compare True alkaloids, proto alkaloids and pseudo alkaloids with examples in each class. 10
2. Write a pharmacognostic note on *Adhatoda*.
- B** Write about sources, chemical constituents with structure, uses, chemical tests and adulterants of *Zer Kachuro*. 10
- C** Write chemistry, isolation and pharmacology of Atropine and Reserpine. 10

Q.4 Attempt any TWO of the following:

- A** Comment on following: 10
1. Nicotine is considered as best insecticide.
2. Kurchi bark is recurved in shape.
3. Vasaka is contraindicated during pregnancy.
4. Pilocarpine is used in ophthalmic surgery.
- B** Write a note on Opium alkaloids. Explain the method of preparation of opium. 10
- C** 1. Enlist various solanaceous plants of your syllabus. Explain chemical constituents and uses of Henbane and belladonna. 10
2. Match the following:
- | | |
|----------------|--------------------|
| a. Ma huang | i. Pseudo alkaloid |
| b. Vinca | ii. Kurchi |
| c. Cola | iii. Anti cancer |
| d. Hollarrhena | iv. Betel nut |
| e. Areca | v. Kutaj |